

RESEARCH INTEREST

- Designing materials and their interfaces for cost-effective, reliable, and high-performance batteries
- Implementing data-driven experimentation workflows to accelerate materials discovery
- Li-ion batteries, Na-ion batteries, aqueous batteries, all-solid-state batteries

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) Mar. 2019 - Aug. 2026 (Expected)
Undergraduate Program Daejeon, South Korea

- Primary Major: **Chemical and Biomolecular Engineering**
- Double Major: **Materials Science and Engineering**
- Overall GPA: **4.21/4.3** (3.983 on 4.0 scale) — CBE GPA: 4.19/4.3, MSE GPA: 4.3/4.3
- Leave of Absence (Fall 2022 - Spring 2024) — fulfilled mandatory military service

Georgia Institute of Technology Aug. 2024 - May 2025
Non-Degree Exchange Program Atlanta, Georgia, U.S.A.

- Overall GPA: **4.0/4.0**
- Relevant Courses: Electrochem Storage, Electrochemical Engr (Grad-Level), Data Analytics Chem Eng

RESEARCH EXPERIENCE

Energy Materials Lab, Dept. of CBE, KAIST Jul. 2025 - Present
Undergraduate Research Assistant (Prof. Nam-Soon Choi)

Project: Discovering Electrolyte Additives for Low-Temperature Na-ion Batteries

- Evaluated representative functional groups through low-temperature capacity retention tests
- Establishing surrogate model based on experimental performance data for rational additive screening

Liu Resarch Group, School of MSE, Georgia Tech Aug. 2024 - May 2025
Undergraduate Research Assistant (Prof. Meilin Liu)

Project: Surface Modification of Ni/Mn-Based Co-Free Layered Cathode for Li-ion Batteries

- Developed Lithium-Molybdenum-Phosphate (LMP) surface modification to address surface instability
- Proposed dual-mechanism of Li_3PO_4 surface layer formation and Mo doping through XRD analysis
- Demonstrated enhanced long-term performance and 2x increase in specific capacity at 5C-rate

Electrochemical Energy Devices Lab, Dept. of CBE, KAIST Jan. 2021 - Dec. 2021
Undergraduate Research Assistant (Prof. Hee-Tak Kim)

Project 1: Cathode Materials Optimization for Flowless Zn-Br Aqueous Batteries

- Evaluated various ordered mesoporous carbons and catalytic additives to optimize energy efficiency

Project 2: Degradation Analysis of Flowless Electrolytic Zn-Mn Aqueous Batteries

- Implemented electrolyte reservoir to relieve H_2 -induced pressure and prevent cell failure
- Attributed carbon felt cathode passivation to irreversible MnO_2 deposition via CV and EIS analysis

PUBLICATION

1. Y. Ahn, X. Hu, Y. Ding, C. Kim, Y.-C. Wu, **T. Kim**, M. Liu*, **Synergistic Effects of Entropy Tuning in Niobium-Based Oxide Anode for Fast-Charging Lithium-Ion Batteries.** *Adv. Funct. Mater.* 2025, e095333. <https://doi.org/10.1002/adfm.202509533>

TECHNICAL SKILL

Synthesis & Fabrication	Solid-State Synthesis, Electrode Fabrication (Slurry, Dry Process), Electrolyte Preparation, Coin Cell Assembly
Electrochemical Analysis	Galvanostatic Cycling, Voltammetry (CV, LSV), EIS, GITT, dQ/dV
Materials Characterization	XRD, XPS, Raman, SEM
Software & Programming	Origin, Python (Machine Learning), MATLAB, $\text{\LaTeX}$ • Developed surrogate model to predict CO₂ capture process (Link) (Course project from the <i>Data Analytics for Chemical Engineers</i>)

ACADEMIC PRESENTATION

Poster Presentations

- 12th Annual MRS Poster Competition**, *MRS Chapter at Georgia Tech* Apr. 2025
- 19th UROP Spring Symposium**, *Undergraduate Research Opportunities, Georgia Tech*
- *Title*: Effect of LMP Surface Coatings on Co-Free Layered Oxide Cathode for Li-ion Batteries ([Link](#))

TEACHING EXPERIENCE

- MAS.10002 Calculus II Tutor**, *School of Freshman, KAIST* Mar. 2020 - Jun. 2022
- Led weekly concept reviews and problem-solving sessions for 5 semesters (Spring 2020 - Spring 2022)

HONOR & SCHOLARSHIP

- KAIST Presidential Fellowship (KPF)**, *KAIST* Mar. 2021 - Present
- Highly selective honor for KAIST undergraduates with outstanding academic excellence
- National Presidential Science Scholarship**, *Korea Student Aid Foundation* Mar. 2021 - Dec. 2024
- Prestigious scholarship for Korean students with exceptional potential in STEM fields
- KOR-US STEM Student Exchange Scholarship**, *Ministry of Trade and Industry* Apr. 2024
- Financial support during Georgia Tech exchange (USD 9,000)
- Merit-Based Academic Excellence Scholarship**, *Dept. of CBE, KAIST*
- *Awarded semesters*: Fall 2020 (2nd place), Spring 2021 (1st place), Fall 2021 (1st place)
- Dean's List**, *College of Engineering, KAIST*
- *Awarded semesters*: Fall 2019, Spring 2020, Fall 2020, Spring 2021, Fall 2021, Spring 2022

EXTRACURRICULAR ACTIVITY

- International House Resident**, *Georgia Tech* Jan. 2025 - May 2025
- Engaged with diverse cultures and strengthened global communication skills
- Social Service Personnel**, *Republic of Korea Army* Nov. 2022 - Aug. 2024
- Alternative mandatory military service; Served at local social welfare center
 - Received *Excellent Social Service Personnel Award* for exemplary conduct (Dec. 2023)
- Departmental Student Council**, *Dept. of CBE, KAIST* Sep. 2020 - Dec. 2021
- Designed and executed virtual photo contest to foster community bonding during pandemic
- Online IT Volunteer**, *World Friends Korea* Nov. 2020
- Designed and delivered a month-long online Adobe Photoshop course for Ugandan students